

Strong Artificial Intelligence

J Language

Behaviour Maker

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Repetition „Mind“

- Why is our mind a time machine?
- What is a thought?
- What is the smallest possible thought?
- Why needs reasoning a logic?
- What are the main tasks of a mind?
- Try to describe what curiosity is.

Workshop „Language“

- What is a language?
- What is a language used for?
- What different kind of languages do you know?
- What criteria can be used to differentiate languages?
- What is the purpose of language?
- What is syntax, grammar and semantics related to languages?

J Language

1. Types

Natural and Artificial

2. Purpose

Action Control and Learning not by Doing

3. Syntax

Structure

4. Grammar

Rules

5. Semantics

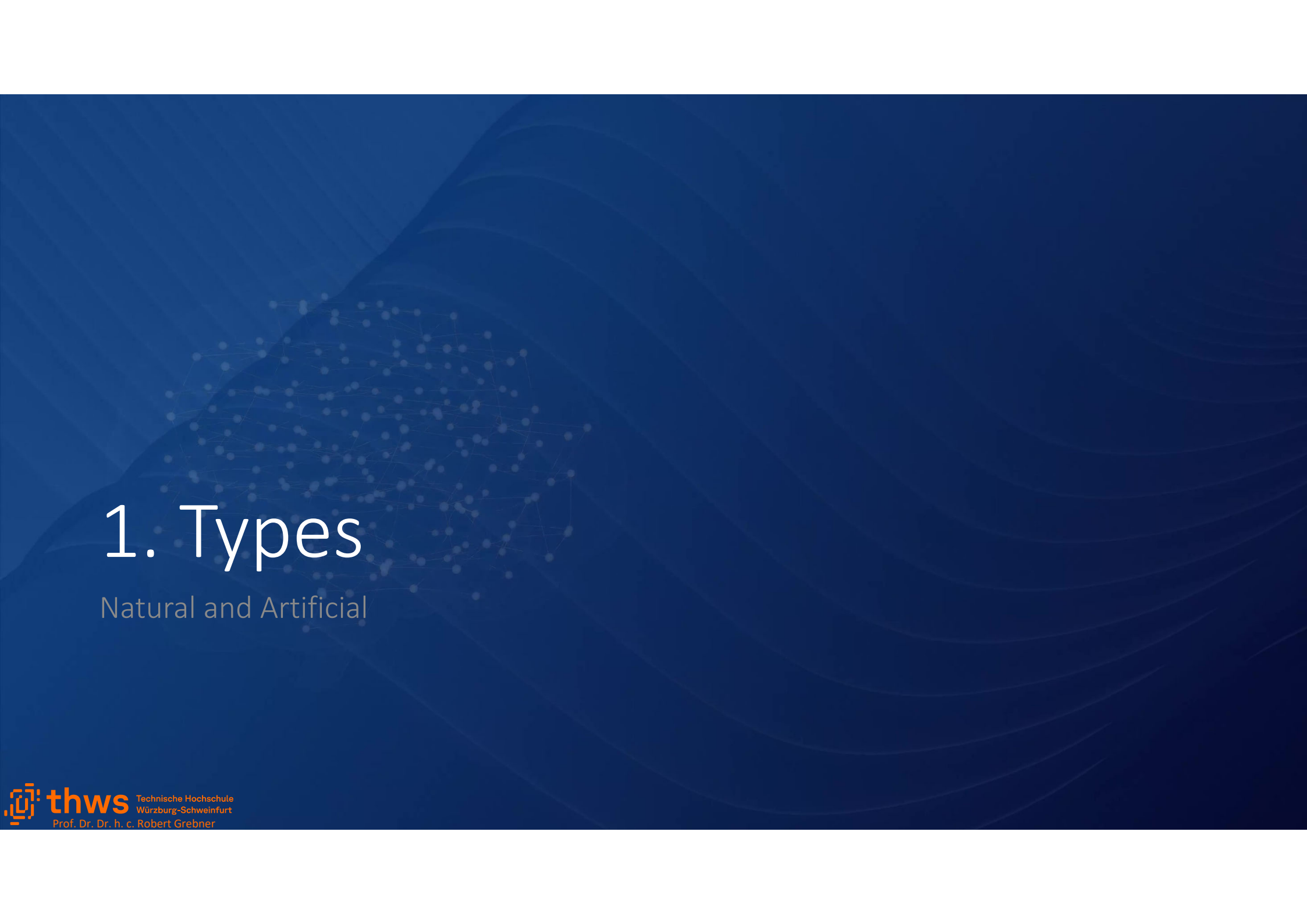
Meaning

6. Brain

Thought Machine

7. Conclusion

Work Empowerer



1. Types

Natural and Artificial

1. Types

Natural Languages

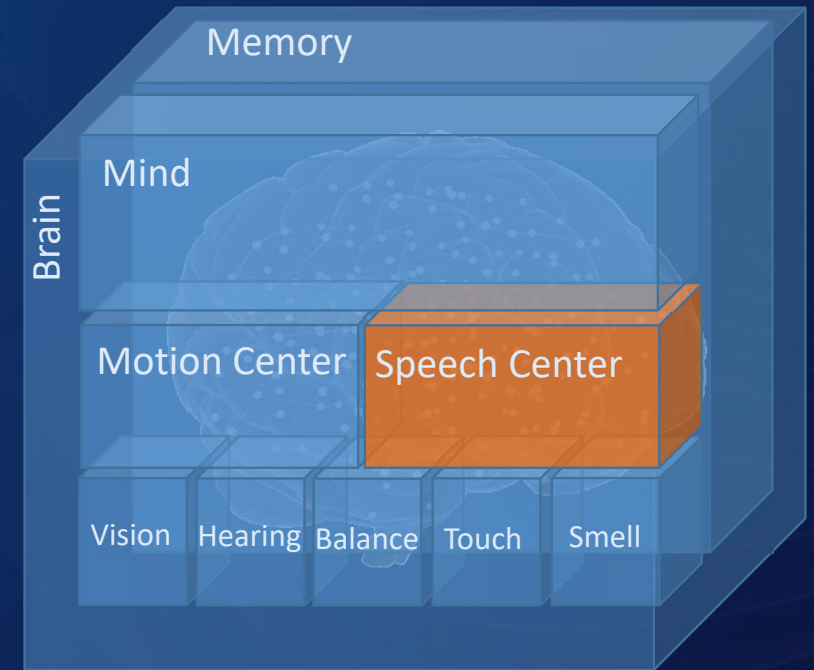
- Human languages
- Animal languages

Artificial Languages

- Formal languages (Programming languages, Modelling languages, Semantic networks)
- Human languages (e. g. Esperanto)

Thought languages (just mental)?

- Is there a thought language?
- Mental entities and their associations?



1. Types

Expression (motion based):

- Oral (speaking)
- Body Language
- Facial Expressions
- Sign language (for the deaf)
- Pantomime
- Writing/Painting/Signs

Medium:

- Sound waves (phonetic on air)
- Photon-based (optically/visual)
- Materials (persistent/repeatable)

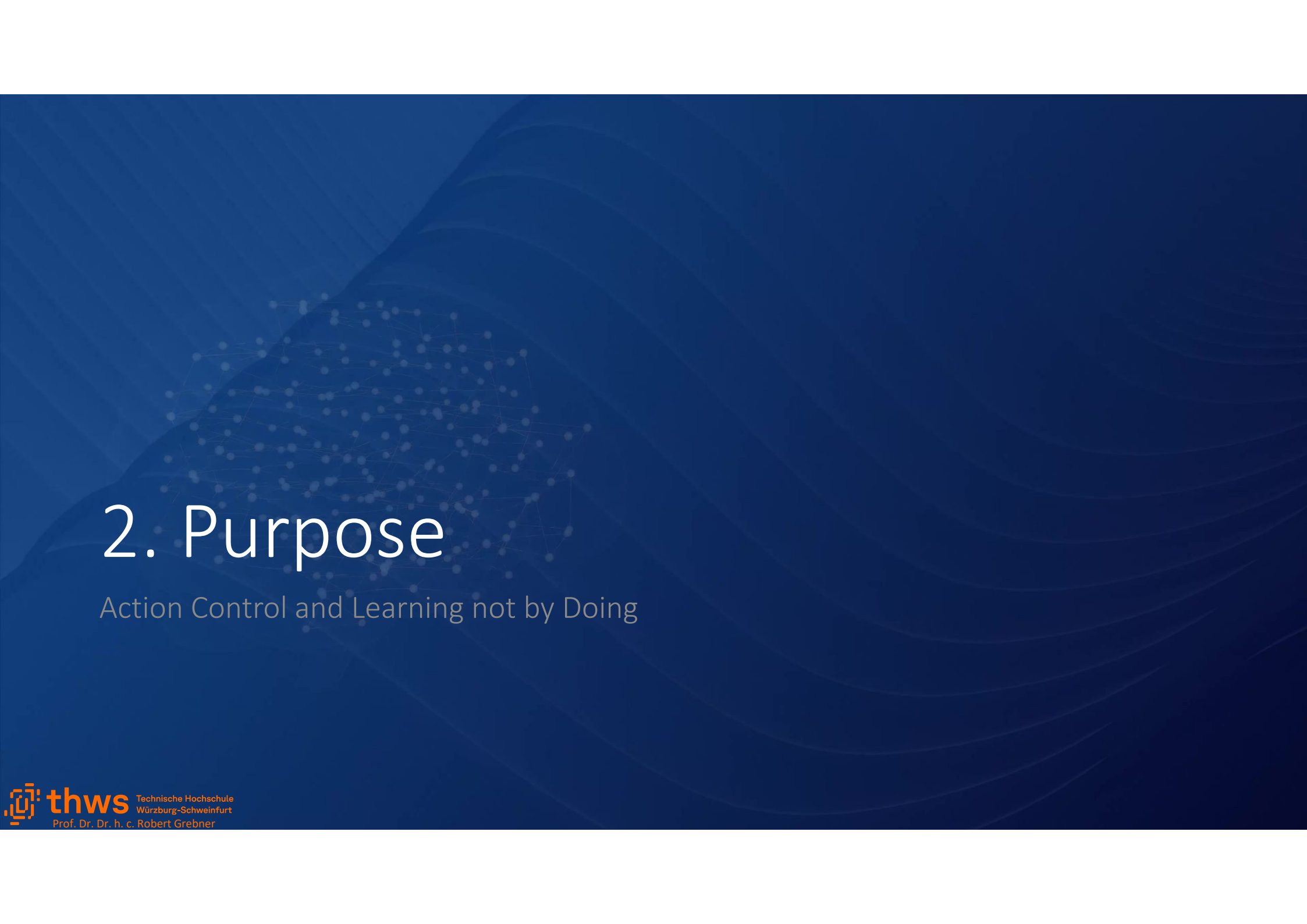
Impression/Perception/Senses

- auditory
- visual
- tactile (e. g. Braille)

Written form

- Phonetic/alphabetic scripts (Latin alphabet)
- Syllabic scripts (e. g. Japanese)
- Consonantal scripts (e. g. Arabic, Hebrew)
- Word and syllabic scripts
Logogrammes (e. g. Chinese, old Egyptian, e. a.)
Signs and symbols (e. g. road signs)
- Picture stories



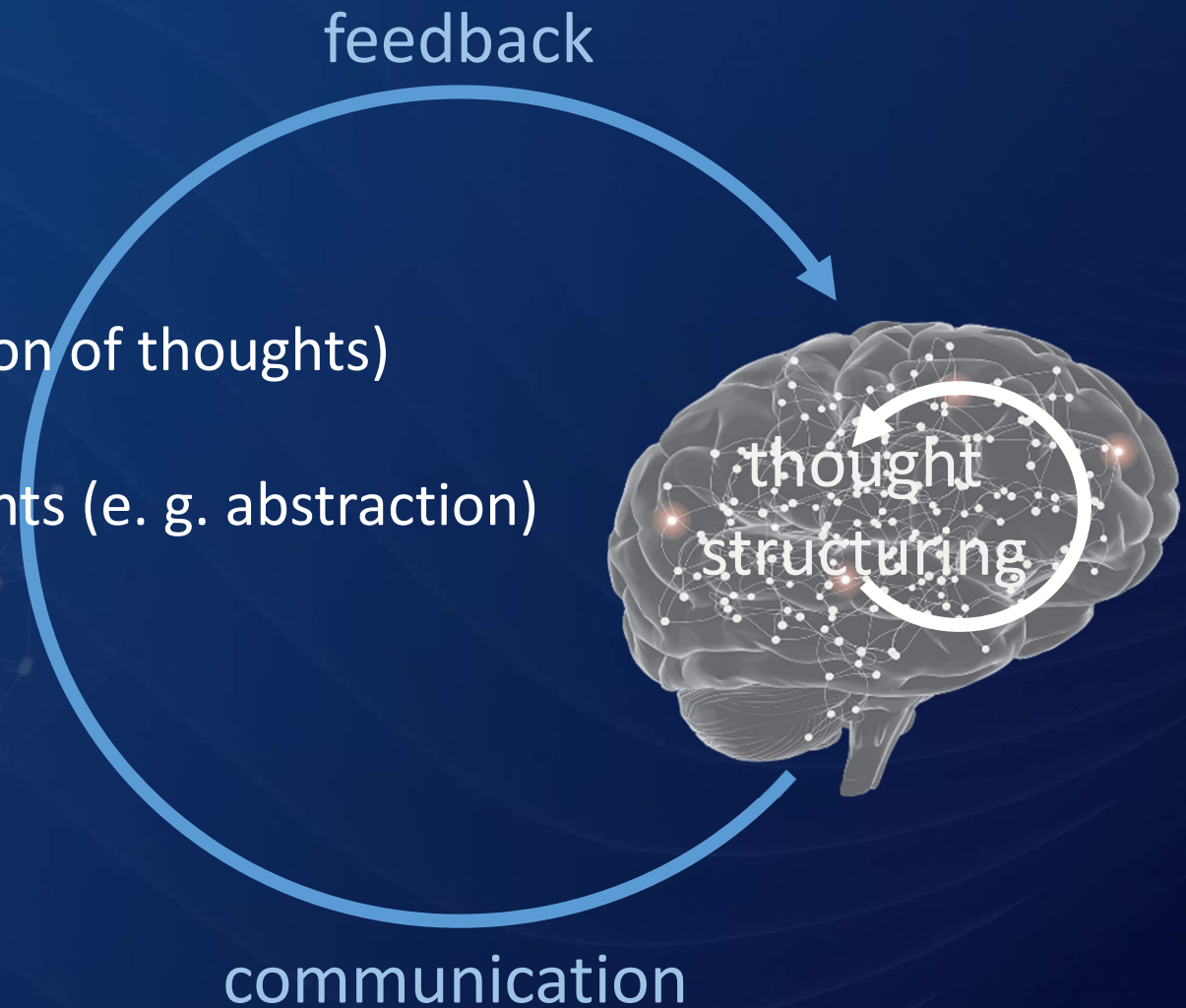


2. Purpose

Action Control and Learning not by Doing

2. Purpose

1. Communication (externalization of thoughts)
2. Support for structuring thoughts (e. g. abstraction)
3. Influencing the way we think?
(feedback loop/logic training)



2. Purpose

Communication: Natural Language



2. Purpose

Communication Artificial/Formal Language

Moving Translated and Materialised Information
(Programme)



```
14 public class Memory {
15
16     // memory segments
17     static public Memory conceptMemory; // memory where the learned concepts are stored, also called individual
18     static public Memory episodicMemory; // memory where all impressions are stored, also called primary memory
19     static public Memory relationalMemory; // memory where all values are stored
20     static public Memory linguisticMemory; // memory where linguistic knowledge is stored
21
22     static public HashMap<Entity, Memory> individualMemory = new HashMap<Entity, Memory> (); // memory representation
23
24     private static Entity myself; // representation of myself
25
26     // basic rules
27     private static Rule ruleOfEnglish;
28     private static Rule ruleOfMath;
29     private static Rule ruleOfPhysics;
30     private static Rule ruleOfChemistry;
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32     // private static Rule ruleOfBiology;
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2. Purpose

Language as a Tool

WHAT (IN A TECHNICAL SENSE)?

Language is about the materialization and mentalisation of information according generally established and accepted rules (grammars).

WHY?

- Transporting information from one brain to another in the narrower sense
- Controlling action and behaviour of social-technical systems in the broader sense

WHO?

- Used by intelligent systems like humans and animals.
- Machines start to behave like systems, that are able to use language (e. g. NLP systems).

HOW?

Rule and convention based with syntax, grammar, semantic, following communication rules.

WHAT (IN A WORK SYSTEM)?

Just a tool but the most important one in group work systems, altering and synchronising information and knowledge of individuals. Even educating minds.



Workshop „Semantics“

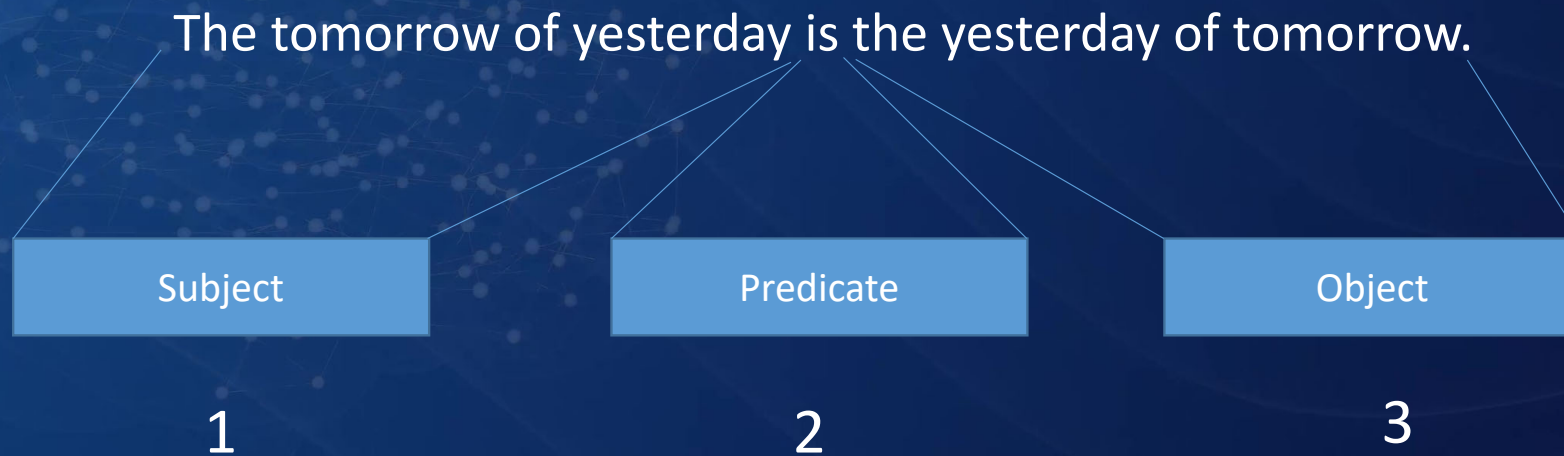
- What is syntax, grammar and semantics and what are the differences or how are they connected to each other?
- Semantic is seen as the meaning of a word, sentence or text. If so, what means the sentence „meaning of a word“?
- What meaning has the sentence „The tomorrow of yesterday is the yesterday of tomorrow“?



3. Syntax

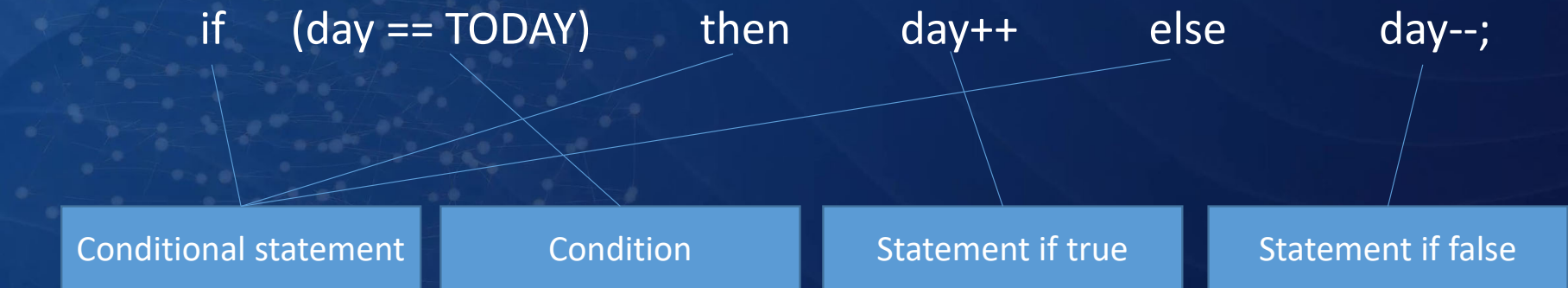
Structure

3. Syntax



A common structure for natural languages is the subject predicate object order

3. Syntax



Typical syntax in procedural or imperative computer languages for a conditional statement



4. Grammar

Rules

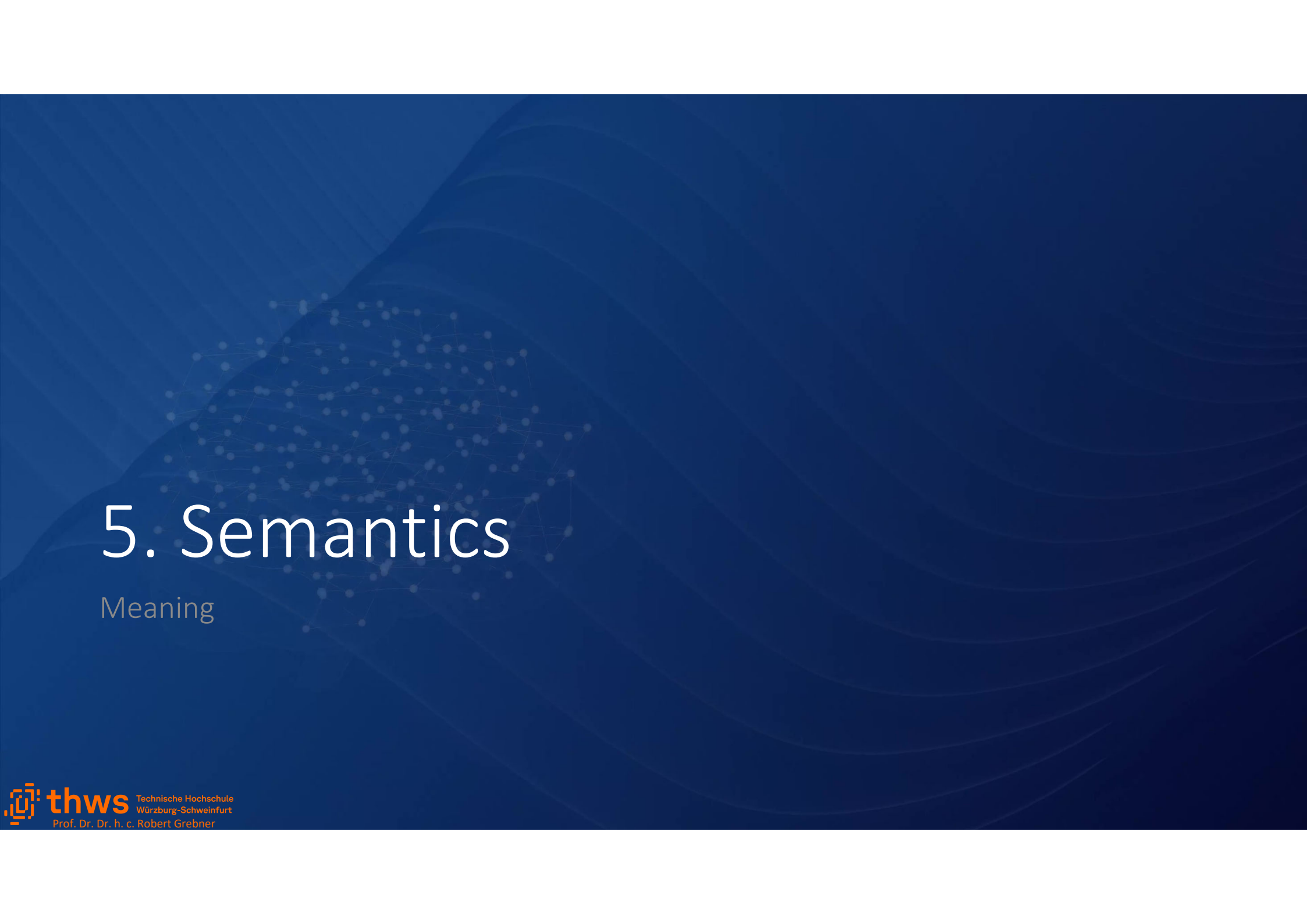
The tomorrows of yesterdays is the yesterdays of tomorrows.

Plural

A common standard for defining the grammar of a (formal) language is the EBNF (extended Bachus-Naur-Form)

`simple declarative sentence ::= subject predicate object`

`conditional statement ::= "if (" condition ")" then" statement
["else" statement] ";"`

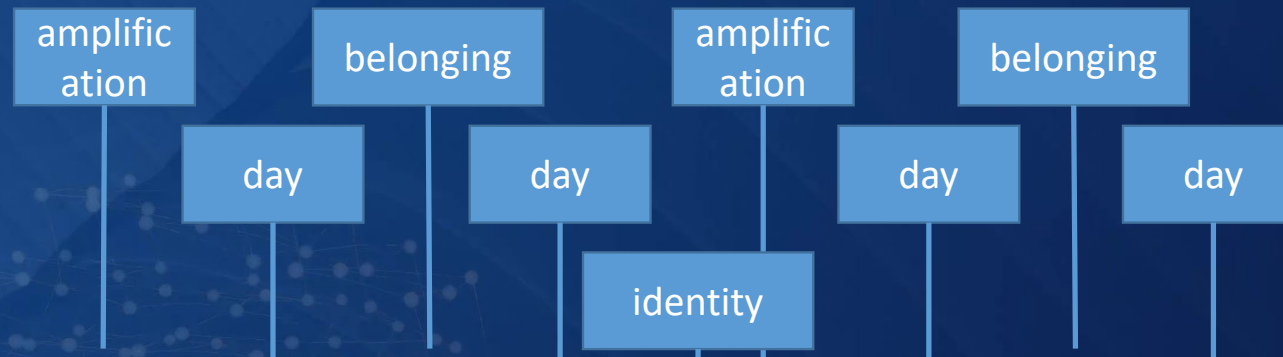


5. Semantics

Meaning

5. Semantics

Meaning



The tomorrow of yesterday is the yesterday of tomorrow.



The tomorrow of yesterday is the yesterday of tomorrow.

Day B: Yesterday means: The day before today (T): $B = T - 1 \text{ day}$

Day A: Tomorrow of Yesterday means: $A = B + 1 \text{ day} = T - 1 \text{ day} + 1 \text{ day} = T$

Day D: Tomorrow means: The day after today (T): $D = T + 1 \text{ day}$

Day C: Yesterday of Tomorrow means: $C = D - 1 \text{ day} = T + 1 \text{ day} - 1 \text{ day} = T$

Meaning of the sentence: $T = T$ or „Today is Today“ (Tautology)

Does this sentence have any meaning at all?

Workshop „Brain“

- What comes to your mind when you think of „day“
- What is the meaning of „day“?
- What is the meaning of „yesterday“ and „tomorrow“?
- What happens in our brain, when we hear the sentence „The tomorrow of yesterday is the yesterday of tomorrow“?
- What is the purpose of that sentence, if it has no meaning?



6. Brain

Meaning Machine



The tomorrow of yesterday is the yesterday of tomorrow.

To understand the meaning of that sentence our brain uses a day logic with the following mental entities:

- Day: All the happenings and movements of entities between the two moments, when the Earth rotates around itself one time.
- Today: the day at which we just get impressions
- Before: observations and impressions we already have in memory, when we get new impressions
- Yesterday: The day before today
- After: observations and impressions we just process are after impressions we already have in memory
- Tomorrow: The day after today
- Time Logic: yesterday (before & after) today; today (before & after) tomorrow

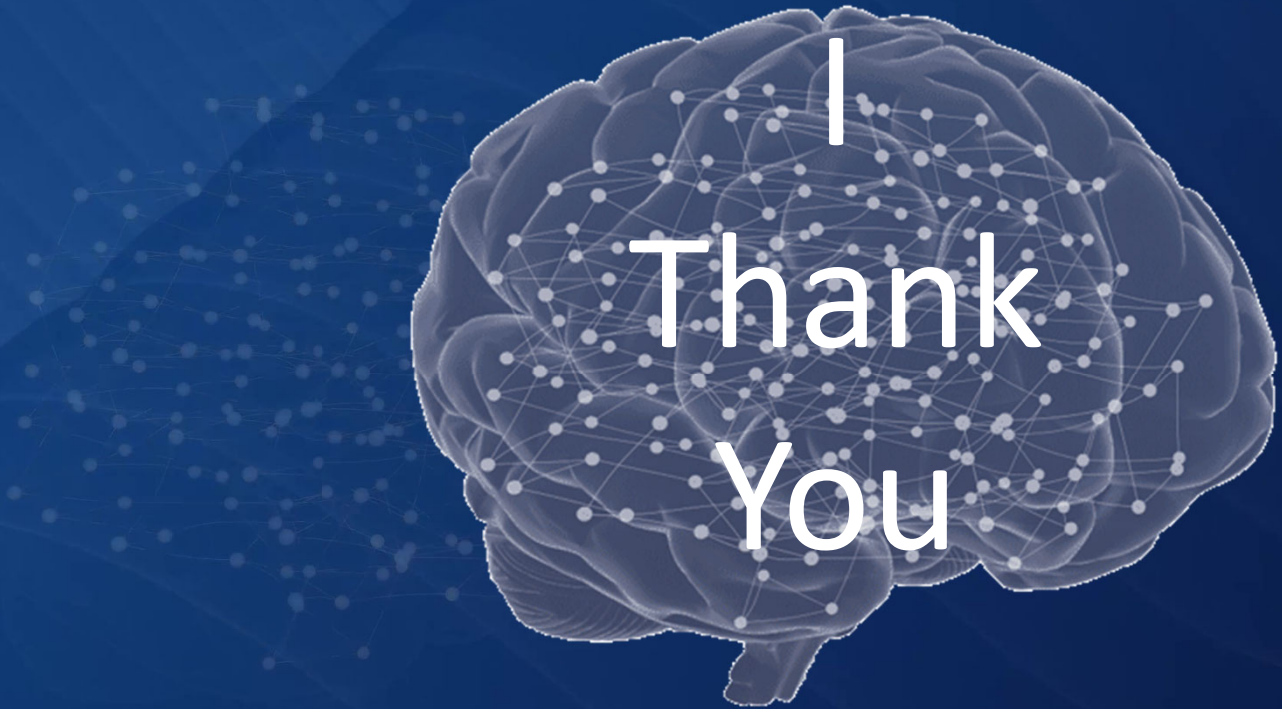
The meaning of a word is what we think or what is intended to be thought, when we hear or read that word (that what is highlighted on our mental or mind's eye).

That means: The word is associated with an known or unknown concept which is a mental entity in our brain.

As long as the meaning is unknown, our brain only remembers the word and normally is starting a learning process when our curiosity is strong enough.

7. Conclusion

- Language is the tool for exchanging information and for controlling systems.
- Perceiving sentences or words immediately starts a mapping process in a brain, to find the right mental entity for the word or the right mental information network for the sentence.
- Semantics is the process of that mapping. In this sense there is a mind processing concept (thinking) behind semantics.



I
Thank
You